Whole Pipeline – Example

6 image → 2 batch × 3 image , 1 epoch → 2 iteration

2 → 2 → 2 (input → hidden → classes)

1, Setup & Data

1.4 Load

$x_1 = [4, 2] \rightarrow 0$, $x_2 = [2, 6] \rightarrow 1$, $x_3 = [5, 1] \rightarrow 0$

$x_4 = [1, 7] \rightarrow 1$, $x_5 = [6, 3] \rightarrow 0$, $x_6 = [0, 5] \rightarrow 1$

$val = [6, 2] \rightarrow 0$, $test = [0, 9] \rightarrow 1$

split → 6 train , 1 val , 1 test

$mean = [\,(4+2+5+1+6+0)/6 , (2+6+1+7+3+5)/6\,] = [3, 4]$

1.5 x − mean

$x_1 = [1, -2]$, $x_2 = [-1, 2]$, $x_3 = [2, -3]$

$x_4 = [-2, 3]$, $x_5 = [3, -1]$, $x_6 = [-3, 1]$

$val = [3, -2]$, $test = [-3, 5]$

2, Model

2.1 affine → batchnorm → ReLU → dropout → affine → softmax

2.2 $dims = [2, 2, 2]$

2.3 $W1 = [\,0.2 , -0.2 ; 0.3 , 0.1\,]$, $b1 = [0, 0]$

$\gamma = [1, 1]$, $\beta = [0, 0]$

$W2 = [\,0.2 , -0.1 ; -0.3 , 0.4\,]$, $b2 = [0, 0]$

$reg = 0.1$, $p = 0.5$

3, Training Setup

3.2 lr = 0.1 , batch = 3 , epoch = 1

3.3 adam : $\beta_1 = 0.9$, $\beta_2 = 0.999$, lr decay = 0.95

iteration = 6 / 3 = 2

m = 0 , v = 0 , t = 0 , best val = 0

running $\mu = [0, 0]$, running $\sigma^2 = [0, 0]$

Iteration 1 ⟳

3.4 batch = {x1 , x2 , x5}

$X = [1 , -2 ; -1 , 2 ; 3 , -1]$, $y = [0, 1, 0]$

4, Forward

4.2 $a = X \cdot W1 + b1$

$[1 \cdot 0.2 + (-2) \cdot 0.3 , 1 \cdot (-0.2) + (-2) \cdot 0.1] = [-0.4 , -0.4]$

$[-1 \cdot 0.2 + 2 \cdot 0.3 , -1 \cdot (-0.2) + 2 \cdot 0.1] = [0.4 , 0.4]$

$[3 \cdot 0.2 + (-1) \cdot 0.3 , 3 \cdot (-0.2) + (-1) \cdot 0.1] = [0.3 , -0.7]$

batchnorm

$\mu = [(-0.4+0.4+0.3)/3 , (-0.4+0.4-0.7)/3] = [0.1 , -0.233]$

$\sigma^2 = [(0.25+0.09+0.04)/3 , (0.028+0.401+0.218)/3] = [0.127 , 0.216]$

$\sqrt{\sigma^2} = [0.356 , 0.464]$

$\hat{x} = (a - \mu) / \sqrt{\sigma^2} = [-1.405 , -0.359 ; 0.843 , 1.364 ; 0.562 , -1.005]$

$\gamma \cdot \hat{x} + \beta = [-1.405 , -0.359 ; 0.843 , 1.364 ; 0.562 , -1.005$

]

running $\mu = 0.9 \cdot 0 + 0.1 \cdot \mu = [0.01, -0.023]$

running $\sigma^2 = 0.9 \cdot 0 + 0.1 \cdot \sigma^2 = [0.013, 0.022]$

ReLU = max(0, ·) = [0 , 0 ; 0.843 , 1.364 ; 0.562 , 0]

dropout : mask = [1 , 1 ; 1 , 0 ; 1 , 1], × mask / 0.5

h = [0 , 0 ; 1.686 , 0 ; 1.124 , 0]

4.3 scores = h · W2 + b2

[0 , 0]

[1.686·0.2 , 1.686·(−0.1)] = [0.337 , −0.169]

[1.124·0.2 , 1.124·(−0.1)] = [0.225 , −0.112]

5, Loss

5.1 scores − max = [0 , 0 ; 0 , −0.506 ; 0 , −0.337]

exp = [1 , 1 ; 1 , 0.603 ; 1 , 0.714]

p = exp / Σ = [0.5 , 0.5 ; 0.624 , 0.376 ; 0.583 , 0.417]

5.2 p correct = [0.5 , 0.376 , 0.583]

5.3 data loss = −(ln 0.5 + ln 0.376 + ln 0.583) / 3

= (0.693 + 0.978 + 0.540) / 3 = 0.737

5.4 reg loss = 0.5 · 0.1 · (Σ W1² + Σ W2²)

= 0.05 · (0.18 + 0.30) = 0.024

5.5 L = 0.737 + 0.024 = 0.761

dout = (p − y_onehot) / 3

= [−0.167 , 0.167 ; 0.208 , −0.208 ; −0.139 , 0.139]

6, Backward

6.2 $dW2 = h^T \cdot dout + reg \cdot W2$

row 1 : $1.686 \cdot 0.208 + 1.124 \cdot (-0.139) + 0.02 = 0.215$

$-0.195 - 0.01 = -0.205$

row 2 : $0 + reg \cdot W2 = [-0.03, 0.04]$

$dW2 = [0.215, -0.205; -0.03, 0.04]$

$db2 = \Sigma\ dout = [-0.098, 0.098]$

$dh = dout \cdot W2^T = [-0.05, 0.117; 0.062, -0.146; -0.042, 0.097]$

6.3 dropout : × mask / 0.5

$= [-0.1, 0.233; 0.125, 0; -0.083, 0.194]$

ReLU : × (out > 0)

$= [0, 0; 0.125, 0; -0.083, 0]$

batchnorm

$d\beta = \Sigma = [0.125 - 0.083, 0] = [0.041, 0]$

$d\gamma = \Sigma\ d \cdot \hat{x} = [0.125 \cdot 0.843 + (-0.083) \cdot 0.562, 0] = [0.058, 0]$

$da = (1/N)/\sqrt{\sigma^2} \cdot (N \cdot d\hat{x} - \Sigma d\hat{x} - \hat{x} \cdot \Sigma (d\hat{x} \cdot \hat{x}))$

$(1/3)/0.356 = 0.936$

r1 : $0.936 \cdot (0 - 0.041 + 1.405 \cdot 0.058) = 0.038$

r2 : $0.936 \cdot (0.375 - 0.041 - 0.843 \cdot 0.058) = 0.266$

r3 : $0.936 \cdot (-0.25 - 0.041 - 0.562 \cdot 0.058) = -0.304$

$da = [0.038, 0; 0.266, 0; -0.304, 0]$

affine

$dW1 = X^T \cdot da + reg \cdot W1$

$[0.038 - 0.266 - 0.912 + 0.02, -0.02] = [-1.12,$

$-0.02]$

$[-0.076 + 0.532 + 0.304 + 0.03 , 0.01] = [0.79 , 0.01]$

$db1 = \Sigma\, da = [0 , 0]$

6.4 grads : $W1 , b1 , \gamma , \beta , W2 , b2$

7, Update (adam) , t = 1

$m = 0.1 \cdot dw , v = 0.001 \cdot dw^2$

$\hat{m} = m / (1 - 0.9) = dw , \hat{v} = v / (1 - 0.999) = dw^2$

$step = lr \cdot \hat{m} / \sqrt{\hat{v}} = 0.1 \cdot sign(dw)$

$W1 = [0.2 , -0.2 ; 0.3 , 0.1] - 0.1 \cdot [-1 , -1 ; 1 , 1] = [0.3 , -0.1 ; 0.2 , 0]$

$b1 = [0 , 0]$

$\gamma = [1 , 1] - 0.1 \cdot [1 , 0] = [0.9 , 1]$

$\beta = [0 , 0] - 0.1 \cdot [1 , 0] = [-0.1 , 0]$

$W2 = [0.2 , -0.1 ; -0.3 , 0.4] - 0.1 \cdot [1 , -1 ; -1 , 1] = [0.1 , 0 ; -0.2 , 0.3]$

$b2 = [0 , 0] - 0.1 \cdot [-1 , 1] = [0.1 , -0.1]$

Check accuracy (first iteration , test mode)

$\hat{x} = (a - \text{running } \mu) / \sqrt{(\text{running } \sigma^2)}$, no dropout

train scores = $[0.1 , -0.1 ; -0.006 , 0.152 ; 0.1 , -0.1 ; -0.204 , 0.356 ; 0.642 , -0.1 ; -0.34 , 0.561]$

argmax = $[0, 1, 0, 1, 0, 1] \rightarrow 6/6 = 1.0$

val : $a = [0.5 , -0.3] \rightarrow \hat{x} = [4.352 , -1.884]$

$\rightarrow [3.817 , -1.884] \rightarrow ReLU\,[3.817 , 0] \rightarrow s = [0.482 , -0.1]$

argmax = $0 \rightarrow 1/1 = 1.0$

1.0 > 0 → best = params of iteration 1

Iteration 2 ⟳

3.4 batch = {x3 , x4 , x6}

$X = [2, -3; -2, 3; -3, 1]$, $y = [0, 1, 1]$

4, Forward

$a = X \cdot W1 = [0, -0.2; 0, 0.2; -0.7, 0.3]$

$\mu = [-0.233, 0.1]$, $\sigma^2 = [0.109, 0.047]$, $\sqrt{\sigma^2} = [0.330, 0.216]$

$\hat{x} = [0.707, -1.389; 0.707, 0.463; -1.414, 0.926]$

$\gamma \cdot \hat{x} + \beta = [0.536, -1.389; 0.536, 0.463; -1.373, 0.926]$

running $\mu = 0.9 \cdot [0.01, -0.023] + 0.1 \cdot \mu = [-0.014, -0.011]$

running $\sigma^2 = 0.9 \cdot [0.013, 0.022] + 0.1 \cdot \sigma^2 = [0.022, 0.024]$

ReLU = [0.536 , 0 ; 0.536 , 0.463 ; 0 , 0.926]

mask = [1 , 1 ; 0 , 1 ; 1 , 1] → h = [1.073 , 0 ; 0 , 0.926 ; 0 , 1.851]

scores = [0.207 , -0.1 ; -0.085 , 0.178 ; -0.270 , 0.455]

5, Loss

p = [0.576 , 0.424 ; 0.435 , 0.565 ; 0.326 , 0.674]

p correct = [0.576 , 0.565 , 0.674]

data loss = (0.552 + 0.571 + 0.395) / 3 = 0.505

reg loss = 0.014 → L = 0.519 (0.761 → 0.519)

dout = [-0.141 , 0.141 ; 0.145 , -0.145 ; 0.109 , -0.109]

6, Backward

dW2 = [-0.142 , 0.152 ; 0.315 , -0.305] , db2 = [0.112 , -0.112]

$d\gamma = [-0.020, -0.168]$, $d\beta = [-0.028, -0.254]$

$dW1 = [-0.124, -0.010; 0.251, -0.447]$, $db1 = [0, 0]$

7, Update (adam), t = 2

$W1[1,1]$: $m = 0.9 \cdot (-0.112) + 0.1 \cdot (-0.124) = -0.113$

$v = 0.999 \cdot 0.00125 + 0.001 \cdot 0.0154 = 0.00127$

$\hat{m} = -0.113 / (1 - 0.9^2) = -0.595$

$\hat{v} = 0.00127 / (1 - 0.999^2) = 0.633$

$W = 0.3 - 0.1 \cdot (-0.595) / \sqrt{0.633} = 0.3 + 0.075 = 0.375$

$W1 = [0.375, -0.007; 0.114, 0.073]$, $b1 = [0, 0]$

$\gamma = [0.861, 1.074]$, $\beta = [-0.113, 0.074]$

$W2 = [0.085, 0.010; -0.268, 0.365]$, $b2 = [0.088, -0.088]$

End of epoch

epoch = 1, lr = 0.1 × 0.95 = 0.095

train argmax = [0, 1, 0, 1, 0, 1] → 1.0, val → 1.0

1.0 > 1.0 ? no → best = params of iteration 1

loop end → model = best

$W1 = [0.3, -0.1; 0.2, 0]$, $\gamma = [0.9, 1]$, $\beta = [-0.1, 0]$

$W2 = [0.1, 0; -0.2, 0.3]$, $b2 = [0.1, -0.1]$

running $\mu = [-0.014, -0.011]$, running $\sigma^2 = [0.022, 0.024]$

8, Evaluation

8.1 train acc = [1.0, 1.0], val acc = [1.0, 1.0]